ME8930: Data-driven Learning and Control of Dynamic Systems: Assignment 3

Course GitHub: `https://github.com/DyCo-AI/data_driven_learning_course`

This assignment studies *control design using data-driven models*. In Assignment 2, you constructed Koopman models using DMD and EDMD methods for nonlinear systems and evaluated their prediction accuracy. In this assignment, you will use those learned linear predictors to design controllers and compare their performance with classical linear controllers designed from local linearization.
For each problem:

- build a linear predictor from data,
- design an LQR controller using the learned data-driven model,
- design an MPC controller using the learned data-driven model,
- compare closed-loop performance between LQR and MPC.

You may use the code in the `mpc` folder of the course repository as a starting point.

Problem 1. Duffing oscillator with control

Consider the controlled Duffing oscillator

$$\dot{x}_1 = x_2,$$
$$\dot{x}_2 = x_1 - \delta x_2 - x_1^3 + u,$$

where $\delta = 0.5$ and $u \in \mathbb{R}$ is the control input.
The control objective is to stabilize the system to the equilibrium point

$$x_{\text{ref}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

1. Verify that $(1, 0)$ is an equilibrium point of the uncontrolled system.
2. Linearize the controlled system about $(1, 0)$ and obtain a local linear model in deviation coordinates. Discretize the model using sampling time $\Delta t = 0.1$ s.
3. Modify the Duffing system to generate training data for the controlled system by simulating trajectories with bounded inputs u_k. Clearly describe how the control inputs are chosen during data collection.
4. Using the controlled training data, construct a lifted linear predictor of the form

$$z_{k+1} = A_z z_k + B_z u_k, \qquad \hat{x}_k = C z_k$$

   using EDMD with a monomial basis. Clearly specify the observables used.

5. Design a discrete-time LQR controller using the local linearized model.

6. Design a finite-horizon MPC controller using the learned lifted model.

7. Simulate the closed-loop nonlinear Duffing system using both controllers for the following initial conditions:

$$\left\{ \begin{bmatrix} 1.15 \\ 0 \end{bmatrix}, \begin{bmatrix} 0.85 \\ 0 \end{bmatrix}, \begin{bmatrix} 1.25 \\ 0.35 \end{bmatrix}, \begin{bmatrix} 0.75 \\ -0.35 \end{bmatrix}, \begin{bmatrix} 1.45 \\ -0.70 \end{bmatrix} \right\}.$$

8. For each initial condition, plot phase portraits (x-y trajectory), state trajectories (vs time), and control inputs (vs time), and briefly discuss how performance changes as the initial condition moves farther from the equilibrium.

9. Compare the closed-loop performance of LQR and MPC using settling time, control effort, maximum absolute input, and terminal state error. Summarize the results in a table.

Problem 2. Pendulum with control

Consider the controlled damped pendulum

$$\ddot{\theta} + b\dot{\theta} + \frac{g}{L}\sin\theta = u,$$

where $b = 2$, $L = 1$, and $g = 9.81$.
Let the state be

$$x = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}.$$

In this problem, you will study two control tasks.

1. Write the pendulum dynamics in first-order state-space form.

2. Modify the pendulum system to generate training data for the controlled system by simulating trajectories with bounded control inputs u_k. Clearly describe how the control inputs are chosen during data collection.

3. Using the controlled training data, construct a lifted linear predictor of the form

$$z_{k+1} = A_z z_k + B_z u_k, \qquad \hat{x}_k = C z_k$$

   using EDMD with a monomial basis. Clearly specify the observables used.

4. For each of the following tasks, do the following:

   - linearize the system about the corresponding equilibrium,
   - discretize the linearized model using sampling time $\Delta t = 0.1$ s,
   - design a discrete-time LQR controller,
   - design a finite-horizon MPC controller using the learned lifted model,
   - simulate the closed-loop nonlinear system for the two specified initial conditions,
   - compare the performance of LQR and MPC.

5. **Task A: Stabilization about the stable equilibrium**
   Stabilize the pendulum about
   $$x_{\text{ref}} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$
   Use the following initial conditions:
   $$\left\{ \begin{bmatrix} 0.25 \\ 0 \end{bmatrix}, \begin{bmatrix} 1.00 \\ 0.30 \end{bmatrix} \right\}.$$

6. **Task B: Stabilization about the unstable upright equilibrium**
   Stabilize the pendulum about
   $$x_{\text{ref}} = \begin{bmatrix} \pi \\ 0 \end{bmatrix}.$$
   Use the following initial conditions:
   $$\left\{ \begin{bmatrix} \pi + 0.12 \\ 0 \end{bmatrix}, \begin{bmatrix} \pi - 0.75 \\ 0.25 \end{bmatrix} \right\}.$$

7. For both Task A and Task B, and for each initial condition, plot phase portraits (x-y trajectory), state trajectories (vs time), and control inputs (vs time), and briefly discuss how performance changes as the initial condition moves farther from the equilibrium.

8. Compare the closed-loop performance of LQR and MPC using settling time, control effort, maximum absolute input, and terminal state error. Summarize the results in a table.